

GIUSEPPE VULDURARO

CHIEF DATA OFFICER · DATA & AI ENGINEER

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Portfolio with nine case studies: giuvul.github.io

SUMMARY

Chief Data Officer with a background in mathematics, leading data and AI at E23, an airport retail analytics startup. Designed and directed a production data platform on AWS (Python, PostgreSQL, Terraform) serving 535M+ rows across 16 airports, and a multi-agent AI system that runs part of the company's operations. Works across engineering, product and clients: from data modeling and pipelines to requirements, roadmap and stakeholder communication.

EXPERIENCE

Chief Data Officer · E23

2024 – Present

Airport retail analytics startup

- Designed and directed E23's passenger profiling platform (~360k lines of code; Python, FastAPI, PostgreSQL on AWS), serving 535M+ rows across 16 airports to customer-facing dashboards.
- Decoupled data generation from database writes with an exactly-once Parquet pipeline (Celery, PyArrow, S3), so compute scales independently of the database.
- Aligned PostgreSQL LIST + HASH partitioning with how clients query the data: each dashboard request reads a single airport slice instead of the full table.
- Introduced declarative data contracts (43 checks, hourly and on every write) with CloudWatch alerting, turning data quality and governance into an automated gate.
- Built the analytics engineering layer with tested dbt models orchestrated by Dagster, and a DuckDB OLAP engine that reuses the Parquet files the pipeline already writes.
- Codified the AWS infrastructure (ECS, RDS, S3, CloudFront, KMS) in Terraform, with GitHub Actions CI/CD, security scanning and verified rollbacks.
- Built a multi-agent AI operations platform (Claude, MCP) with LLM-as-judge evaluation calibrated against human raters and human approval on every outbound action.
- Worked directly with airport and retail clients: translated business questions into data products and KPIs, and shaped the product roadmap with the team.

SKILLS

Data engineering: Python, SQL, PostgreSQL, ETL / ELT pipelines, Data modeling, Partitioning & query optimization, Apache Spark, Apache Kafka, Parquet / PyArrow, Celery, Redis, DuckDB

Analytics engineering: dbt, Dagster, Data contracts, Data quality, Data governance, Semantic layer, Dashboards & KPIs

Platform & DataOps: AWS (ECS, RDS, S3, CloudFront, CloudWatch, IAM), Terraform, Docker, CI/CD with GitHub Actions, Observability

AI engineering: LLM agents, Multi-agent systems, Claude / Anthropic API, MCP, RAG, Prompt engineering, LLM evaluation, Human-in-the-loop design

Software: FastAPI, Pydantic, SQLAlchemy, Alembic, pytest, Git

Mathematics & machine learning: Statistics, Probability, Linear algebra, Optimization, Mathematical modeling, Classical machine learning

Business & product: Client requirements, Pre-sales, Proposals & pricing, Contract negotiation, Product roadmap, Stakeholder communication, Technical leadership

EDUCATION

**First-level Master in AI and AI Agents for Business (60 ECTS) · Università degli Studi
Guglielmo Marconi**

Jan 2026 – Dec 2026 (expected)

Bachelor's Degree in Mathematics · Sapienza Università di Roma

2022 – 2024

LANGUAGES

Italian (native) · English (C1) · Spanish (conversational)